- They serve as solvent for other fats. Example: Fat soluble vitamins A, D, E and K
- The saturated fatty acids have higher melting points than unsaturated fatty acids of corresponding length.

Common Name	Melting Point
Palmitic acid (Saturated fatty acid)	63° C
Oleic acid (Unsaturated fatty acid)	13° C

Note

The trans-double bond isomer of oleic acid, known as elaidic acid, has a linear shape and a melting point of 45° C (32° C higher than its cis isomer).

Chemical properties

i. Hydrolysis:

Triglycerides (fats) can be hydrolyzed to produce glycerol and fatty acids in the presence of acid and heat or with a suitable lipase enzyme under biological conditions.

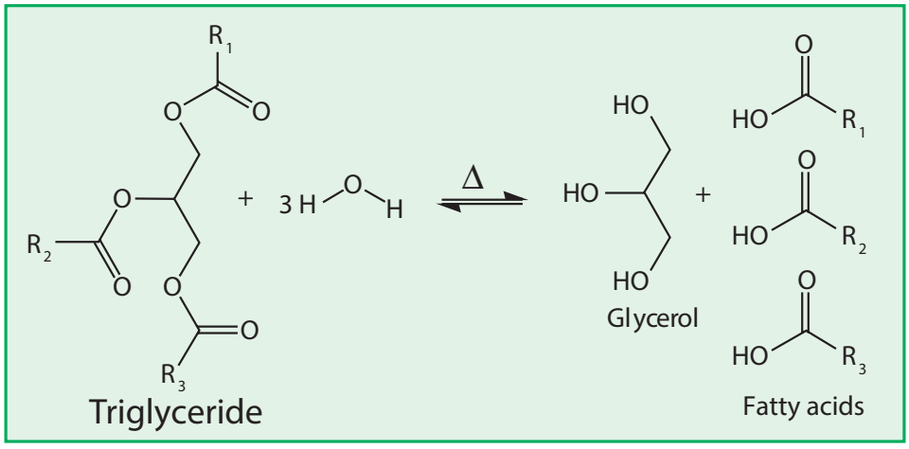


Figure 6.8 Hydrolysis of triglycerides

ii. Hydrogenation:

Hydrogenation is a process of adding hydrogen atoms to unsaturated fats until they become saturated. Hydrogenation of fat is a process used in industries, food manufacturers, to synthesize modified plant fats called hydrogenated fats that share similar texture and taste characteristics with saturated animal fats.

Note

During the processing, many fatty acids saturate and then spontaneously convert back to a unsaturated state, but in a trans-isomer form.

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